5.1)

a) $x(t) = ut(1 - 2\sin\omega t)$

con $u = 2\ m/s$ $\omega = 4\ s^{-1}$

$x(t) = 0$? → $ut(1-2\sin\omega t)$

$t = 0$ ↓ $1 - 2\sin\omega t = 0$

$\sin\omega t = \frac{1}{2}$ → $\sin\omega t = \frac{1}{2}$

$\begin{cases} t = \frac{\pi}{6} + 2k\pi \\ t = \frac{5\pi}{6} + 2k\pi \end{cases}$

$\bar{v}(t) = \frac{\Delta x}{\Delta t} = \frac{(7,54 - (-1,79))}{0,5} = 17,54\ m/s$

$\begin{cases} t = \frac{\pi}{6\cdot 4} + 2k\pi \\ t = \frac{5\pi}{6\cdot 4} + 2k\pi \\ t = 0 \end{cases}$ ← $x(t) = 0$

$x(t) = ut - 2ut\sin\omega t$

$v(t) = u - 2u(\sin(\omega t) + \omega t\cos(\omega t))$

$= u - 2u\sin(\omega t) - 2u\omega t\cos(\omega t)$

$v(0,75) = 19,97\ m/s$

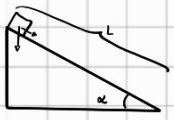
c) $a(t) = -2u\cdot\omega\cos(\omega t) + 2u\omega^2 t\sin(\omega t)$

$= 2u\omega(\omega t\sin(\omega t) - \cos(\omega t))$

$a\left(\frac{\pi}{6\cdot 4}\right) = -14,5\ m/s^2$

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m scivola lungo $\overline{AO}$ di lunghezza L



$\mu_0 = \frac{1}{2}$

ma varia A seconda della legge $\mu_0(s) = \mu_0 - ks^2$ (attenti: potenza) dove s è la distanza percorsa. $\tan\alpha = \frac{3}{4}$

P è partito da fermo in A

$v_{0\,finale}$?

$E_{mec,i} = U_g + K$

$= mgh + \frac{1}{2}mv_0^2$

$E_{mec,fin} = U_g + K$

$= mgh + \frac{1}{2}mv_f^2$

$W_{NS} = F_{NS}\cdot s$

$= \mu_0(s)\cdot N\cdot s$

$= (\mu_0 - ks^2)\cdot s\cdot N = (\mu_0 - ks^2)\cdot s\cdot P\cdot\cos(\alpha)$

$N = P\cdot\cos(\alpha)$

$\Delta E_{mec} = W_{NS} = \frac{1}{2}mv_f^2 - mgh = (\mu_0 - ks^2)\cdot s\cdot P\cdot\cos(\alpha)$

$s = L$

$W_A = -\int_0^L \mu_0(s)\cdot m\cdot g\cdot\cos\alpha\ ds =$ $\mu_0(L) = \frac{1}{2} - kL^2 = 0$ ⇓ $k = \frac{1}{2L^2}$

$= -mg\cdot\cos\alpha\cdot\int_0^L \mu_0 - ks^2\ ds =$

$= -mg\cos\alpha\left[\mu_0 s - k\frac{s^3}{3}\right]_0^L =$

$= -Lmg\cos\alpha\left(\mu_0 - k\frac{L^2}{3}\right) =$ $\mu_0 = \frac{2}{6} = \frac{1}{3}$

$= -Lmg\cos\alpha\left(\frac{1}{2} - \frac{1}{6}\right) = -\frac{4}{15}mgL$

$W_P = mgh$

$= mg\cdot\sin\alpha\cdot L = \frac{3}{5}mgL$

$\Delta K = W_{TOT}$

$\frac{1}{2}mv_f^2 = W_P + W_A = \frac{9}{15}mgL - \frac{4}{15}mgL =$

$\frac{1}{2}mv_f^2 = \frac{5}{15}mgL$

$v_f = \sqrt{\frac{2}{3}gL}$

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3. $m = 5\ kg$

$A = (0; 0,5)\ m$

$B = (2; 0,5)\ m$

$F_{max} = 20\ N$

W: dalle forze agenti sul corpo

$W_{peso} = mg\cdot h = 5\cdot 9,81\cdot 0,5 = -14,715\ J$

$W_{Forza} = F(\Delta x) = 40\ J$ $\quad W_{tot} = 25,29\ J$

b) v_B? $v_A = 0$

$W_{tot} = \Delta K = K_B - K_A$

$= \frac{1}{2}mv_B^2$

$v_B = \sqrt{\frac{2W_{tot}}{m}} = 3,18\ m/s$

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1. $\vec{F}(x,y) = (\alpha x + \beta y,\ \gamma x + \delta y)$

è conservativa se $\vec{F} = 0$ e U potenziale

$\vec{F}(P,Q)$ definito in tutto $\mathbb{R}$ è conservativo quando è irrotazionale le derivate parziali incrociate devono essere uguali:

$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

$\boxed{\beta = \gamma}$ è la condizione affinché sia conservativo

$\vec{F} = -\nabla U$

$\frac{\partial U}{\partial x} = -(\alpha x + \beta y)$

$\frac{\partial U}{\partial y} = -(\beta x + \delta y)$

$U(x,y) = -\int(\alpha x + \beta y)\ dx = -\frac{\alpha x^2}{2} - \beta yx + g(y)$

$\frac{\partial U}{y} = -\beta x + g'(y) = -\beta x - \delta y$

$g'(y) = -\delta y$

$g(y) = \int -\delta y\ dy = -\frac{1}{2}\delta y^2 + C$

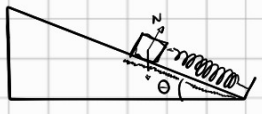
$U(x,y) = -\frac{1}{2}\alpha x^2 - \beta yx - \frac{1}{2}\delta y^2 + C$

$m = 5\ kg$

$K = 9,8\ N/m$

$\mu_s = 0,4$

$\mu_0 = 0,1$



$\theta_{max} = 22°$ $\quad mg\sin\theta \le \mu_s\cdot N = \mu_s\cdot\cos\theta\cdot mg$ ← $F_{att,max}$

$\frac{\sin\theta}{\cos\theta} = \mu_s = \tan(\theta)$

$\tan^{-1}(\mu_s) = \theta_{max} = 21,8°$

$\theta = 22°$ fisso → $ma_x = -kx^2 + mg\sin\theta - \mu_0 N\cdot\Delta x$

$\begin{cases} ma_x = -F_{el} + P_x - F_{att} \\ ma_y = 0 \end{cases}$ ↓ $a_x = 0$

$-kx^2 + \mu_0 N\cdot\Delta x + mg\sin\theta = 0$

$\Delta = b^2 - 4ac = \mu_0^2 N^2 + 4\cdot k\cdot mg\sin\theta = 7203,24$

$\sqrt{\Delta} = 84,87$

$x_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-4,55 \pm 84,87}{-19,6}$ → $0,456\ m$ → $45,6\ cm$

$\theta = \frac{\pi}{6}$

massa M

lunghezza L

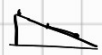


$v_0 \neq 0$

$\mu_0 = \frac{2}{\sqrt{3}}$

$v_{0,min}$ = ? che permette al pendolo di compiere un giro completo

$\cos\theta = \frac{\sqrt{3}}{2}$



$F_\perp = M\cdot g\cdot\cos\theta$

$F_{Attr} = F_\perp\cdot\mu_0 = Mg\cos\theta\cdot\mu_0$

$W_{Attrito} = -F_{attr}\cdot s = -Mg\cdot\cos\theta\cdot\mu_0\cdot 2\pi L$

s = distanza di un cerchio di raggio L

$= 2\pi L$

Nel punto più alto ≥ 0

$E_1 = \frac{1}{2}Mv_0^2$

$E_{finale} = Mg(2L\sin\theta) + \frac{1}{2}Mv_{min}^2$

$v_{min} > 0$ $\quad \frac{1}{2}Mv_0^2 + W_{attrito} = Mg(2L\sin\theta)$

$\frac{1}{2}Mv_0^2 - Mg\cos\theta\,\mu_0 2\pi L = Mg2L\sin\theta$

$v_0 = \sqrt{2gL(2\sin\theta + \mu_0 2\pi\cos\theta)}$

$\frac{1}{2}$ $\quad \frac{2}{\sqrt{3}}$ $\quad \frac{\sqrt{3}}{2}$

$= \sqrt{2g(1 + 2\pi)}$

$K = 50\ N/m$ $\quad m = 0,05\ kg$ $\quad$ no gravità

L non importa se a segno

$P_1(L,0,0)$

$P_2(0,L,0)$

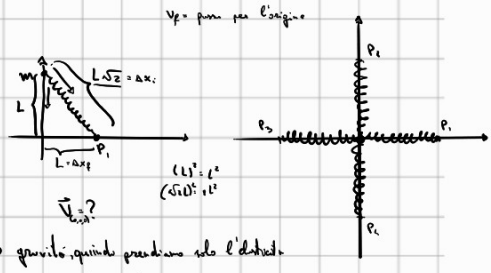
$P_3(-L,0,0)$

$P_4(0,-L,0)$

$L = 0,2\ m$

a) $P(0,0,L)$ con $v_0 = 0$

v_f = passa per l'origine



$\vec{v}_{in,0}$?

No gravità, quindi prendiamo solo l'elastico dell'intero sistema

$K_i + U_{el,i} = K_f + U_{el,f}$

$\frac{1}{2}mv_i^2 + 2k\Delta x_i^2 = \frac{1}{2}mv_f^2 + 2k\Delta x_f^2$

$\sqrt{\frac{4kL^2}{m}} = v_f$

$v_f = 2L\sqrt{\frac{k}{m}} = 12,65\ m/s$

b) $\frac{d^2z}{dt^2} = -\omega^2 z$

$-4kz = ma_z$

$-4kz = m\frac{d^2z}{dt^2} \Rightarrow \frac{d^2z}{dt^2} = -\frac{4k}{m}z$

$\omega = \sqrt{\frac{4k}{m}} = 2\omega = 2\sqrt{\frac{k}{m}}$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{2}\sqrt{\frac{1}{k}} = 0,0445$

$\approx 0,1\ s$

c) $E_{mec} = K + U_{el,1} + U_{el,2} + U_{el,3} + U_{el,4}$

$= \frac{1}{2}\left(mv^2 + k(\delta_1^2 + \delta_2^2 + \delta_3^2 + \delta_4^2)\right)$

$\delta_1 = \sqrt{(x-L)^2 + y^2}$ $\quad \delta_3 = \sqrt{(x+L)^2 + y^2}$

$\delta_2 = \sqrt{x^2 + (y-L)^2}$ $\quad \delta_4 = \sqrt{x^2 + (y+L)^2}$

$\frac{1}{2}k\left((x-L)^2 + y^2 + x^2 + (y-L)^2 + (x+L)^2 + y^2 + x^2 + (y+L)^2\right)$

$x^2 + L^2 - 2xL + y^2 + x^2 + L^2 - 2yL + x^2 + L^2 + 2xL + y^2 + x^2 + L^2 + 2yL$

$2x^2 + 2y^2 + 4L^2 + 2x^2 + 2y^2$

$2k\,(x^2+y^2+L^2)$

$r = \sqrt{x^2+y^2}$

$U(r) = 2k(r^2+L^2)$

$r = 2L$

$U(r=2L) = 2k\,(5L^2) = 10kL^2 = \underline{20\,J}$